

Total Effective Carbon Price

Executive Summary

Every climate policy changes fuel prices. Some policies make dirty fuels more expensive relative to clean ones (**fuel switching**). Some raise the overall cost of energy (**demand reduction**). Most do both. These effects vary across sectors and fuels.

The **total effective carbon price** τ^* answers one question: *what single economy-wide carbon tax would reduce emissions by the same amount as the actual policy mix?*

$$\tau^* = \frac{1}{\Lambda} \sum_s \sum_f E_{fs} \left[\underbrace{\tilde{\varepsilon}_{fs}^{\text{int}} \cdot \tau_{fs}^{\text{int}}}_{\text{switching}} + \underbrace{\tilde{\varepsilon}_s^{\text{ext}} \cdot \tau_s^{\text{ext}}}_{\text{demand}} \right]$$

The tilde $\tilde{\varepsilon}$ denotes **semi-elasticities**: the proportional change in emissions per *unit* change in carbon price (\$/tCO₂), not per percentage change. Semi-elasticities are the natural form when the policy variable is an absolute price.

s, f	Sector and fuel indices.
E_{fs}	Baseline emissions from fuel f in sector s (tCO ₂).
$\tilde{\varepsilon}_{fs}^{\text{int}}$	Intensive semi-elasticity : proportional shift in fuel f 's consumption per \$/tCO ₂ of relative price signal. Fuel- and sector-specific. Units: (tCO ₂ /\$) ⁻¹ $\equiv$ per \$/tCO ₂ .
τ_{fs}^{int}	Intensive carbon price on fuel f in sector s : the component of the policy that changes this fuel's price <i>relative to others</i> , in \$/tCO ₂ .
$\tilde{\varepsilon}_s^{\text{ext}}$	Extensive semi-elasticity : proportional reduction in total energy demand per \$/tCO ₂ of average cost increase. Sector-level. Same units.
τ_s^{ext}	Extensive carbon price in sector s : the component that raises the <i>average</i> cost of energy, in \$/tCO ₂ .
Λ	Normalisation constant : the emissions reduction from a \$1/tCO ₂ uniform carbon tax across the whole economy. Depends only on the energy balance, prices, and elasticities—not on the policy. Computed once per country.

Intuition. The numerator sums the emissions reduction from every fuel in every sector, split into two channels. Each channel is a semi-elasticity times a carbon price—the more responsive the economy and the stronger the price signal, the bigger the reduction. The denominator Λ converts the total reduction into “equivalent dollars of carbon tax.”

Key property. If the policy *is* a uniform carbon tax at rate τ , then $\tau_{fs}^{\text{int}} = \tau_s^{\text{ext}} = \tau$ for all f, s , and the formula returns $\tau^* = \tau$ exactly.

Examples. A revenue-neutral feebate has $\tau_s^{\text{ext}} \approx 0$: only the switching channel contributes. A uniform energy tax has $\tau_{fs}^{\text{int}} = 0$: only the demand channel contributes. The effective carbon price reveals how much bang each instrument delivers per dollar, and allows them to be added.

Appendix: Deriving the Semi-Elasticities

A. From isoelastic elasticities to semi-elasticities

An **isoelastic (standard) elasticity** gives the proportional response to a proportional price change:

$$\varepsilon = \frac{\partial \ln q}{\partial \ln p} \implies \frac{\Delta q}{q} \approx \varepsilon \frac{\Delta p}{p}$$

A **semi-elasticity** gives the proportional response to an *absolute* price change:

$$\tilde{\varepsilon} = \frac{\partial \ln q}{\partial p} \implies \frac{\Delta q}{q} \approx \tilde{\varepsilon} \Delta p$$

Since $\partial \ln p = \partial p/p$, the conversion is simply:

$$\boxed{\tilde{\varepsilon} = \frac{\varepsilon}{p}} \tag{1}$$

where p is the price level at which the semi-elasticity is evaluated (the baseline fuel price for the intensive margin, or the CES cost index for the extensive margin).

B. From CES parameters to isoelastic elasticities

Consider a sector s with CES substitution elasticity σ_s and service demand elasticity η_s . Fuel f has energy share $w_{fs} = q_{fs}/Q_s$, emission factor e_f , and price p_{fs} .

Intensive (fuel-switching) elasticity. The CES own-price elasticity of fuel f , holding total sectoral energy constant, is:

$$\varepsilon_{fs}^{\text{int}} = -\sigma_s (1 - w_{fs}) \tag{2}$$

This says: a 1% rise in the price of fuel f reduces its quantity by $\sigma_s(1 - w_{fs})\%$, with the displaced energy redistributed across other fuels. The $(1 - w_{fs})$ term reflects that a fuel with a dominant share has less scope for substitution.

But for the emissions formula we need the elasticity of fuel f 's *emissions* with respect to a carbon price signal (which changes price in proportion to e_f). The effective intensive elasticity acting on emissions, per unit of intensive carbon price, is:

$$\varepsilon_{fs}^{\text{int}} = \sigma_s (1 - w_{fs}) (e_f - \bar{e}_s) \tag{3}$$

where $\bar{e}_s = \sum_f w_{fs} e_f$ is the energy-share-weighted mean emission factor. The $(e_f - \bar{e}_s)$ term captures that fuel switching only reduces emissions to the extent that the fuel being displaced is dirtier than the sector average.

Extensive (demand) elasticity. The elasticity of total sectoral energy demand to the CES cost index is:

$$\varepsilon_s^{\text{ext}} = \eta_s \tag{4}$$

For the emissions formula, a carbon tax of \$1/tCO₂ raises the cost index by approximately $\bar{e}_s$ dollars per GJ, so the effective extensive elasticity acting on emissions is:

$$\varepsilon_s^{\text{ext}} = |\eta_s| \bar{e}_s \tag{5}$$

C. From CES parameters directly to semi-elasticities

Combining sections A and B in one step:

Intensive semi-elasticity (fuel f , sector s):

$$\tilde{\varepsilon}_{fs}^{\text{int}} = \sigma_s (1 - w_{fs}) \frac{e_f - \bar{e}_s}{p_{fs}} \quad (6)$$

- σ_s : CES substitution elasticity (how substitutable are fuels in this sector)
- $(1 - w_{fs})$: room to substitute (smaller share $\Rightarrow$ more room)
- $(e_f - \bar{e}_s)$: carbon intensity relative to sector average
- $1/p_{fs}$: converts from elasticity to semi-elasticity (cheaper fuels are more responsive in absolute terms)

Extensive semi-elasticity (sector s):

$$\tilde{\varepsilon}_s^{\text{ext}} = |\eta_s| \frac{\bar{e}_s}{C_s} \quad (7)$$

- $|\eta_s|$: demand elasticity (how responsive is energy demand to cost)
- $\bar{e}_s$: average carbon intensity of the sector's energy mix
- $1/C_s$: converts from elasticity to semi-elasticity (CES cost index plays the role of price)

Normalisation constant (for the full economy):

$$\Lambda = \sum_s \sum_f E_{fs} \left[\tilde{\varepsilon}_{fs}^{\text{int}} + \tilde{\varepsilon}_s^{\text{ext}} \right] \quad (8)$$

This is the total emissions reduction from a \$1/tCO₂ carbon tax, since under a uniform carbon tax both τ_{fs}^{int} and τ_s^{ext} equal \$1.

Summary of the full chain:

Start from		Arrive at
CES parameters (σ_s, η_s)	$\xrightarrow{\text{Section B}}$	Isoelastic elasticities $(\varepsilon_{fs}^{\text{int}}, \varepsilon_s^{\text{ext}})$
Isoelastic elasticities	$\xrightarrow{\text{Section A: } \div p}$	Semi-elasticities $(\tilde{\varepsilon}_{fs}^{\text{int}}, \tilde{\varepsilon}_s^{\text{ext}})$
Semi-elasticities + policy prices	$\xrightarrow{\text{Main formula}}$	Effective carbon price τ^*